DEPARTMENT OF ELECTRICAL ENGINEERING

COLLEGE OF E&ME, NUST, RAWALPINDI

DEPARTMENT OF ELECTRICAL ENGINEERING

EE-313 Electronic Circuit Design

Project Report
Audio Amplifier

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Objective:

The objective of this project is to design and implement a complete audio amplifier using **only discrete components** (MOSFETs, BJTs, and passive elements). The amplifier must take input from an **electret condenser microphone**, amplify low-level audio signals through a multi-stage design, and deliver **3–5 W output power** to an **8-Ω speaker** with a clear frequency response from **100Hz to 20 kHz**.

The project includes **circuit calculations, LTspice simulations, component selection justification, and stage-wise testing** of all amplifier blocks including:

1. MOSFET pre-amplifier
2. BJT pre-amplifier
3. Combined pre-amplifier with feedback
4. PNP driver stage
5. Class-AB power amplifier output stage

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Introduction:

Audio amplifiers are essential circuits used to boost low-level sound signals to levels suitable for driving loudspeakers. Microphones generate signals in the range of **millivolts**, which require careful amplification with **low noise**, **stable gain**, and **wide bandwidth**.

In this project, a multi-stage discrete audio amplifier is designed according to specifications using **MOSFETs and BJTs instead of op-amps**. Each stage performs a specific function:

- The **preamplifier** increases the microphone's small signal while maintaining high input impedance.
- The **driver stage** prepares the signal for the power amplifier.
- The **Class-AB output amplifier** provides high power, low distortion, and good efficiency to drive a speaker.

Simulations were performed on **LTspice**, and analytical calculations were done separately for each transistor stage (biasing, gain, input/output impedance, and frequency response).

Finally, all stages were combined and tested to meet the required specifications.

Class AB Power Amplifier (Output Stage):

Purpose

- Deliver 3–5 W power to 8 Ω load
- Provide high efficiency
- Reduce crossover distortion
- Allow ±5 V clean output swing

Calculations:

$$V_{CC} = \sqrt{8PR_L} + V_{CEsat} + V_{CEsat} + V_{R2}$$

$$V_{CC} = \sqrt{8PR_L} + 2.5V + 2.5V + 2V$$

$$\therefore R_L = 8\Omega \quad \therefore P = 5Watt$$

$$V_{CC} = \sqrt{8(5)(8)} + 2.5V + 2.5V + 2V$$

$$\boxed{V_{CC} \approx 24V}$$

"For Vout":

Assume $\hat{V}_o = \frac{V_{CC}}{2}$

$$\hat{V}_o = \frac{24}{2}$$

$$\hat{V}_o = 12V$$

$$V_{rms} \approx \frac{\hat{V}_o}{\sqrt{2}}$$

$$\boxed{V_{rms} = 5V}$$

$$I_{out} = \frac{V_{out}}{R_L}$$

$$I_{out} = \frac{5V}{8}$$

$$I_{rms} = 0.625A$$

$$\boxed{\hat{I}_{out} \approx 1.2A}$$

Assume voltage drop acros R_1 is 6V.

$$R_1 = \frac{6}{10 \times I_R}$$

$$R_1 = \frac{6}{10 \times 0.999 \times 10^{-3}}$$

$$\boxed{R_1 = 600\,\Omega}$$

The Quiesent current for Tip 122 & Tip127:

$$I_Q = \frac{V_B - V_{BE}}{R_E}$$

$$I_Q = I_{se}\, e^{\left(\frac{V_{bias}}{2V_T}\right)}$$

$$I_Q = 10^{-12}\, e^{\left(\frac{1.2}{50 \times 10^{-3}}\right)}$$

$$\boxed{I_Q \approx 17.3\,mA}$$

For Q2 :-

Ic from circuit = 24mA

$$i_B = \frac{i_c}{\beta}$$

$$i_B = \frac{24m}{100}$$

$$\boxed{i_B = 2.4 \times 10^{-4} A}$$

R4 and R5 current (I) is 10 times of i_B.

$$I = 10 \times 2.4 \times 10^{-4}$$

$$\boxed{I = 2.4 \times 10^{-3} A}$$

Assume 10 % of voltage drop at emitter.

$$V_E = 2.4V$$

$$R_E = \frac{2.4 - 0}{24m} = 100\,\Omega$$

$\boxed{R_E \approx 150\,\Omega} \Rightarrow$ 120 bypassed, 30 unbypassed.

$$V_B = V_E + 0.7$$

$$V_B = 2.4 + 0.7$$

$$\boxed{V_B = 3.1V}$$

$$R_{4top} = \frac{24 - 3.1}{2.4 \times 10^{-3}} = 8708\,\Omega$$

$$\boxed{R_{4top} \approx 8k\,\Omega}$$

$$R_{5\,bottom} = \frac{3.1 - 0}{2.4 \times 10^{-3}} = 1292\,\Omega$$

$$\boxed{R_{5\,bottom} \approx 1.3k}$$

Rin:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel (\beta+1)R_{E\,(unbypassed)}$$

$$R_{in} = R_4 \parallel R_5 \parallel (\beta+1)R_E$$

$$R_4 \parallel R_5 = \left.\frac{8k \times 1.3k}{8k + 1.3k}\right| \parallel (101)30$$

$$R_{in} = 1118.27 \parallel 3030$$

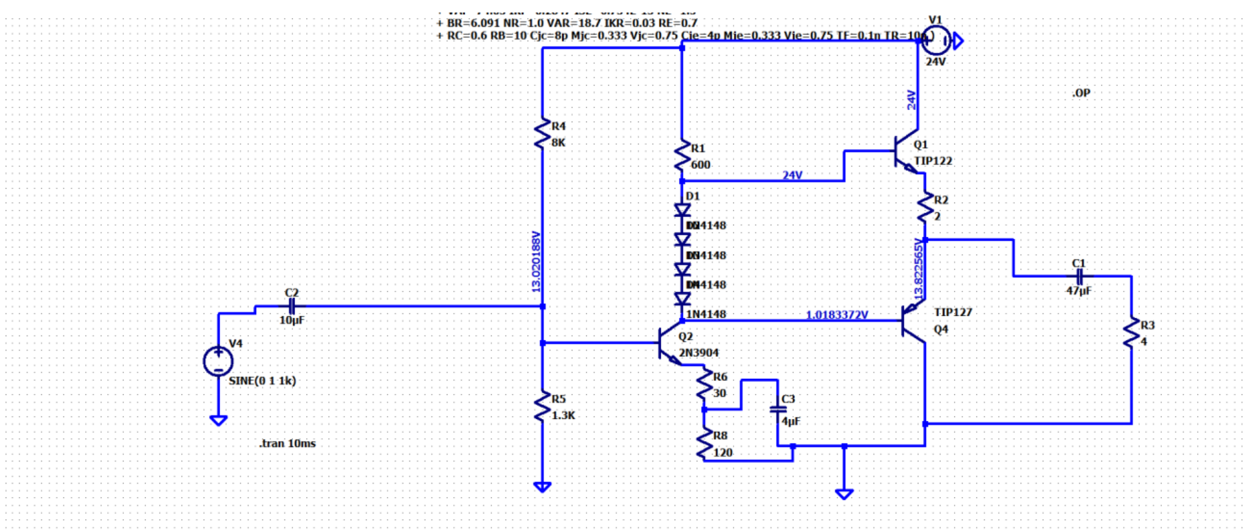
$$R_{in} = \frac{1118.27 \times 3030}{1118.27 + 3030}$$

$$R_{in} = 816.8\,\Omega \approx 820\,\Omega$$

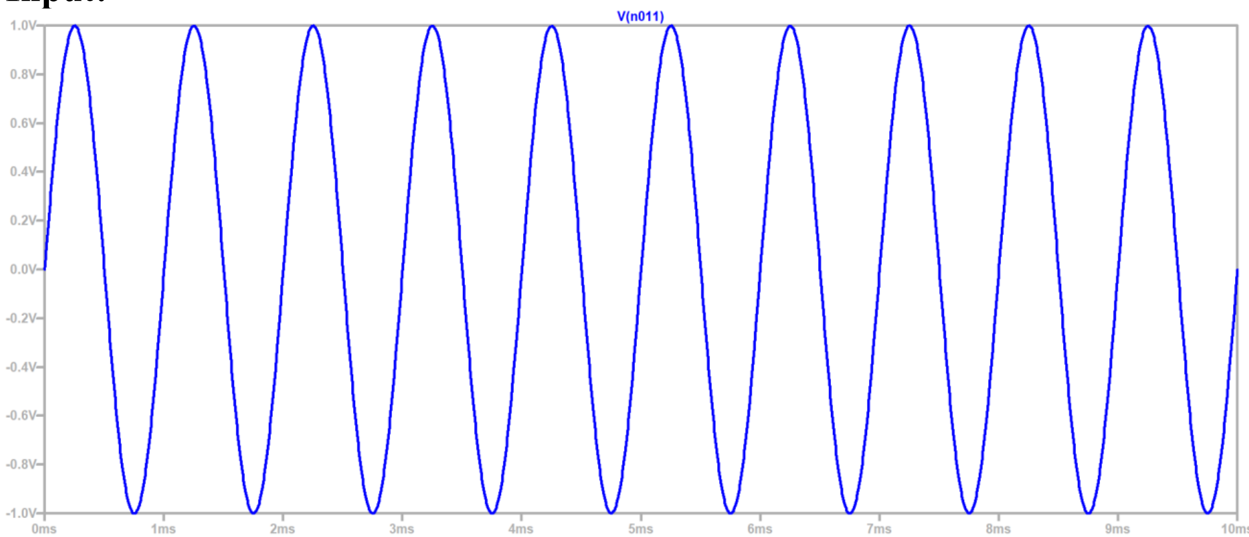
$$\boxed{R_{in} \simeq 820\,\Omega}$$

$\Rightarrow R_L$ for PNP $\simeq 820\,\Omega$

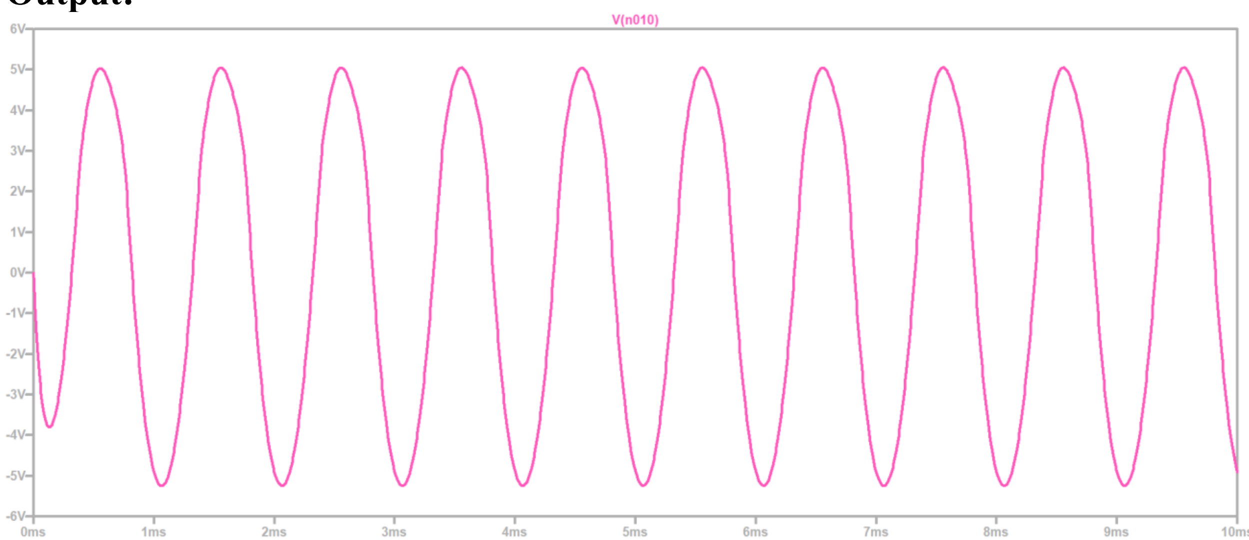
Circuit Diagram (LTspice Schematic):



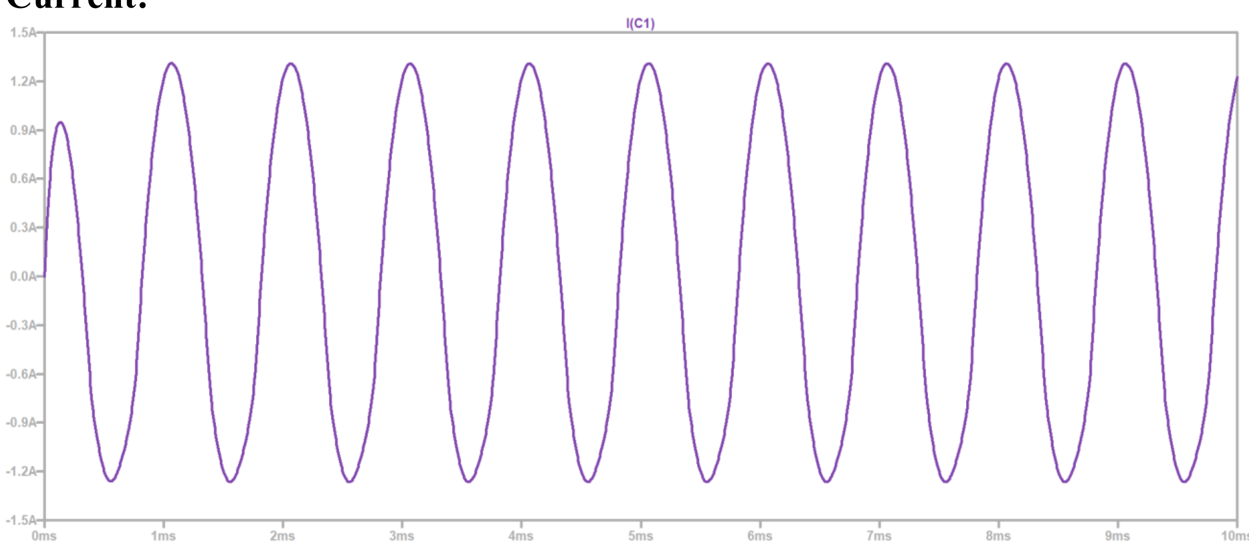
Input:



Output:



Current:



Small Conclusion

"Class-AB stage successfully delivers the required 3–5 W to the load with low distortion. This is the core power stage on which all previous stages depend."

PNP Driver Stage:

Purpose:

- Provide the necessary input current to drive the Class-AB output transistors
- Improve voltage swing
- Prevent loading of preceding stage

Calculations:

For Q3 PNP :-

Assume $I_c = 7mA$

$$V_c = \frac{V_{cc}}{2} = \frac{24}{2} = 12V$$

$$R_c = \frac{24-12}{7mA} = 1714.2\Omega$$

$$R_c = R_7 \approx \boxed{1714\Omega}$$

⇒ Voltage drop across emitter is 10% of Vcc.

$$R_E = \frac{2.4-0}{7mA} = 342.85$$

$$R_E = R_9 \approx \boxed{343\Omega}$$

In PNP:-

$$V_B = V_E - 0.7$$

$$V_B = 2.4 - 0.7$$

$$\boxed{V_B = 1.7V}$$

$$i_B = i_c/\beta$$

$\therefore \beta = 100$

$$i_B = \frac{7m}{100} = \boxed{7\times10^{-5}A}$$

Current 10 times.

$$I = 10 \times 7\times10^{-5}$$

$$\boxed{I = 7\times10^{-4}A}$$

$$R_{B1top} = R_{10} = \frac{24-1.7}{7\times10^{-4}} = 31857\Omega$$

$$\boxed{R_{10} \simeq 2.88k}$$

$$R_{B2bottom} = R_{11} = \frac{1.7-0}{7\times10^{-4}} = 2428.57$$

$$\boxed{R_{11} \simeq 1.92k}$$

Impedence Matching (Between Driver stage and PreAmplifier):

Purpose

- Ensure pre-amplifier does not load the PNP driver
- Ensure maximum signal transfer
- Stabilize gain and frequency response

Small Conclusion:

"Impedance between stages was matched to ensure stable gain, prevent loading, and maintain good bandwidth."

Rin :

Rin of PNP :-

$$R_{in} = R_{B1} \parallel R_{B2} \parallel (\beta+1) R_E \text{ (unbypassed)}$$

$$R_{in} = R_{10} \parallel R_{11} \parallel (101)(50)$$

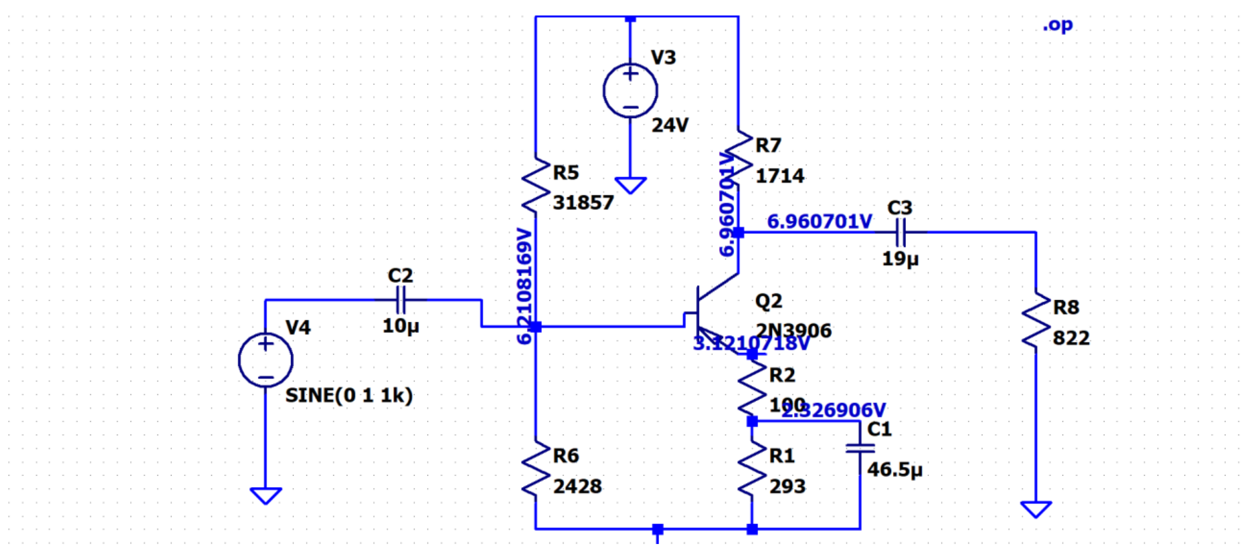
$$= \frac{31857 \times 2428}{31857 + 2428} \parallel (101)(50)$$

$$= 225.11 \parallel 5050$$

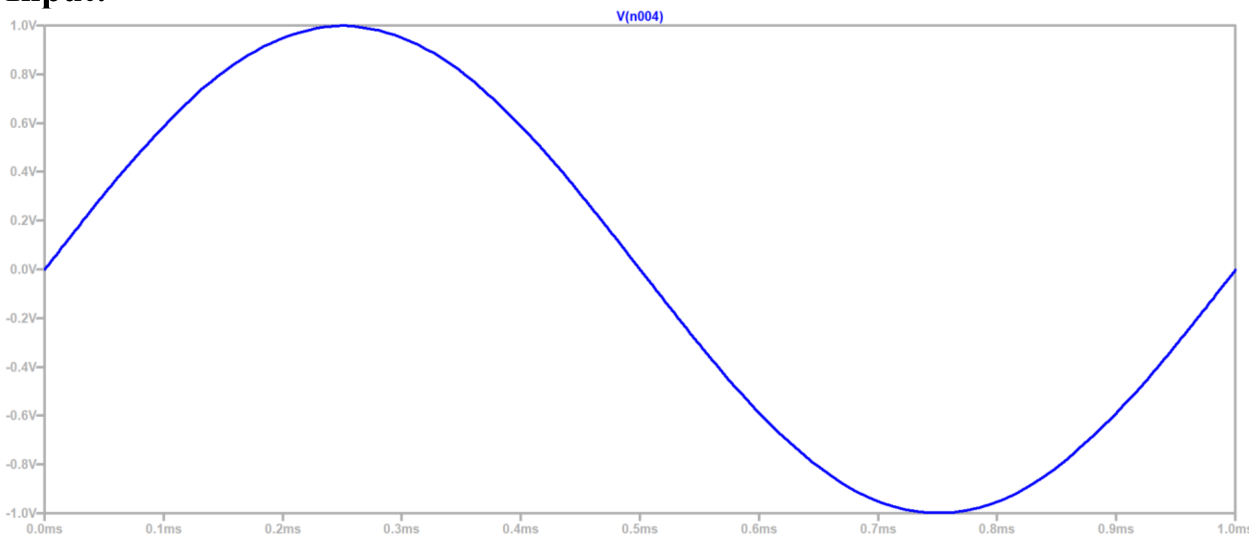
$$\text{Rin of PNP} = \boxed{1559.85\,\Omega}$$

$$\Rightarrow R_L \text{ for prem amp} = 1559.85\,\Omega$$

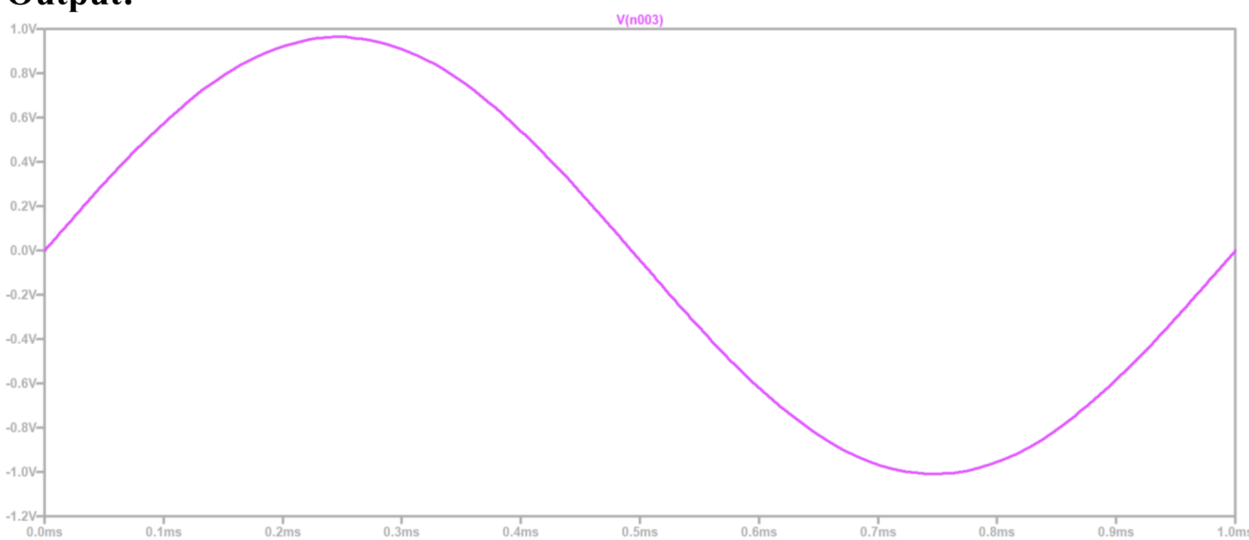
Circuit Diagram (Ltspice Schematic):



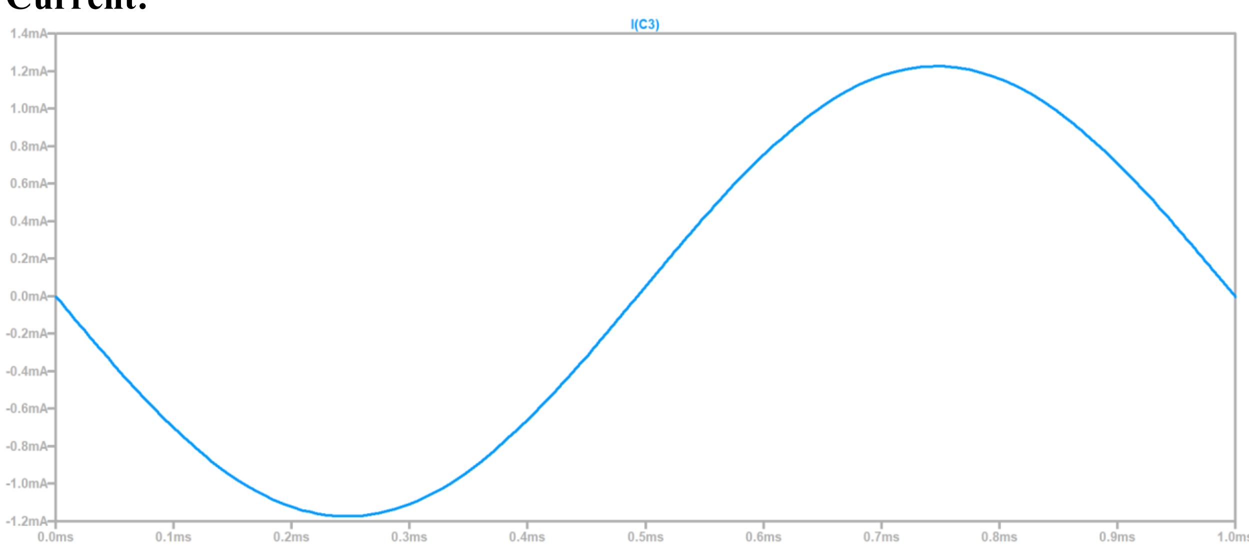
Input:



Output:



Current:



Small Conclusion:

"PNP driver provides sufficient current gain and ensures that the Class-AB stage operates properly without clipping."

Combined Class AB Amplifier:

In the final design, the PNP driver transistor and the Class-AB output transistors together form the complete power stage of the amplifier. Although the PNP driver is part of the Class-AB structure, it was first calculated separately to determine the required bias current, voltage swing, and load conditions. After these calculations, the driver was merged with the Class-AB output pair to create one unified high-power stage.

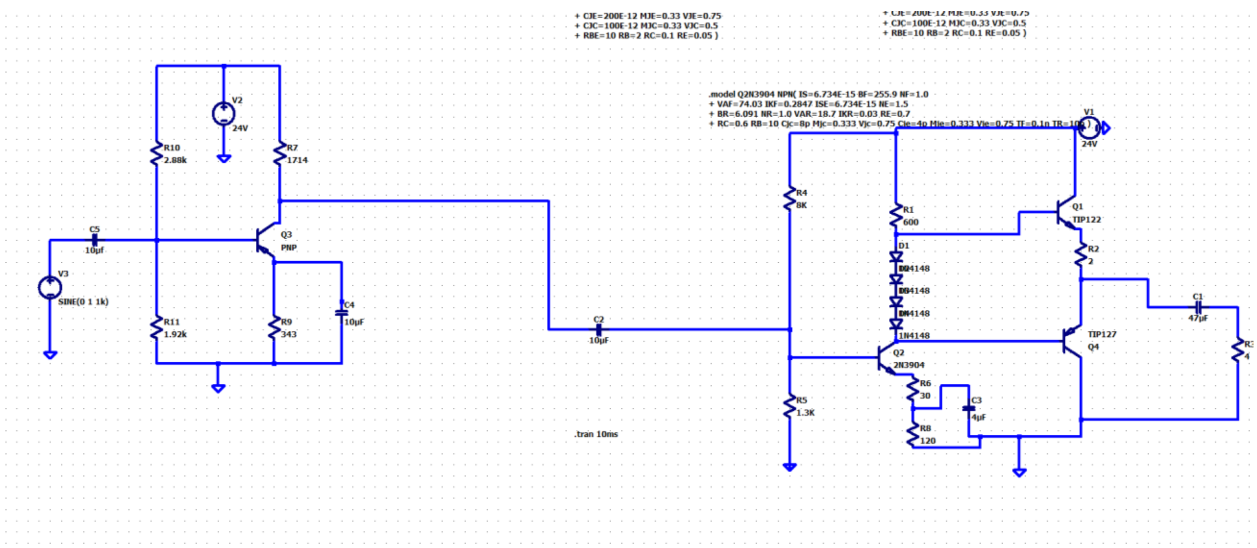
The combined stage works as follows:

- The PNP driver increases the input current and prepares the signal so the output transistors can operate efficiently.
- The Class-AB output transistors then deliver the final high-power signal to the 8-Ω speaker with low distortion.

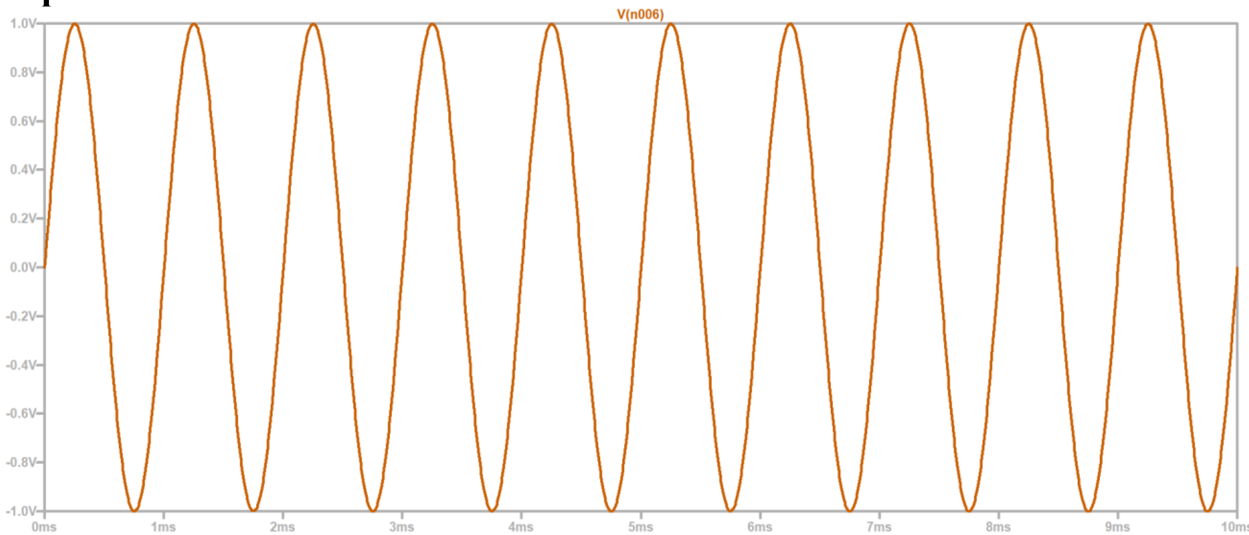
Simulations of the combined block confirmed:

- proper biasing of all transistors,
- clean undistorted voltage swing,
- and successful delivery of 3–5 W output power.

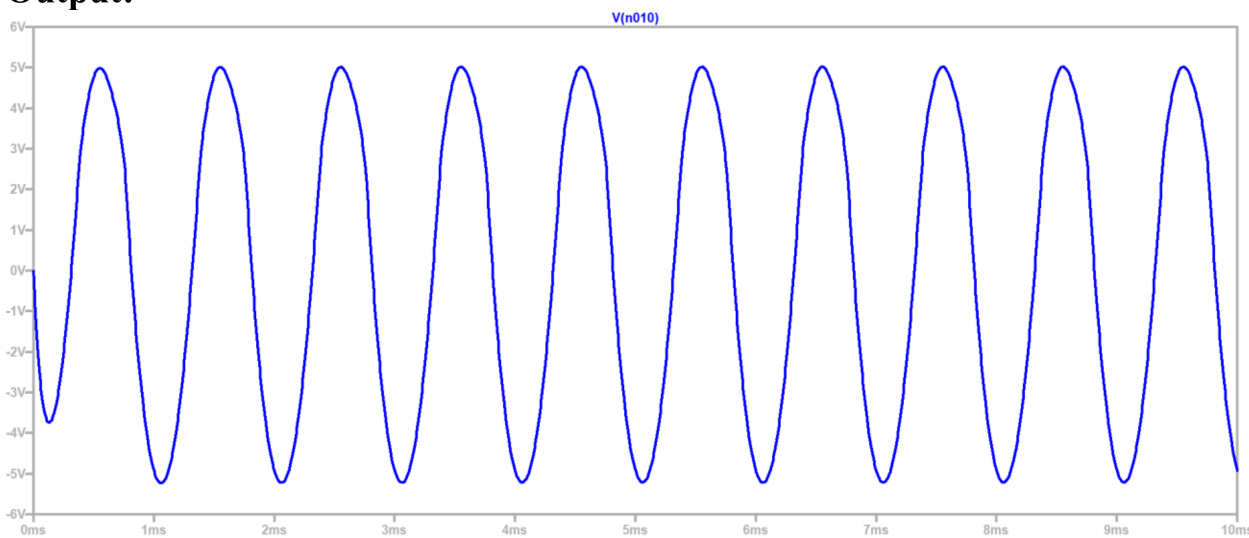
Circuit Diagram (Ltspice Schematic):



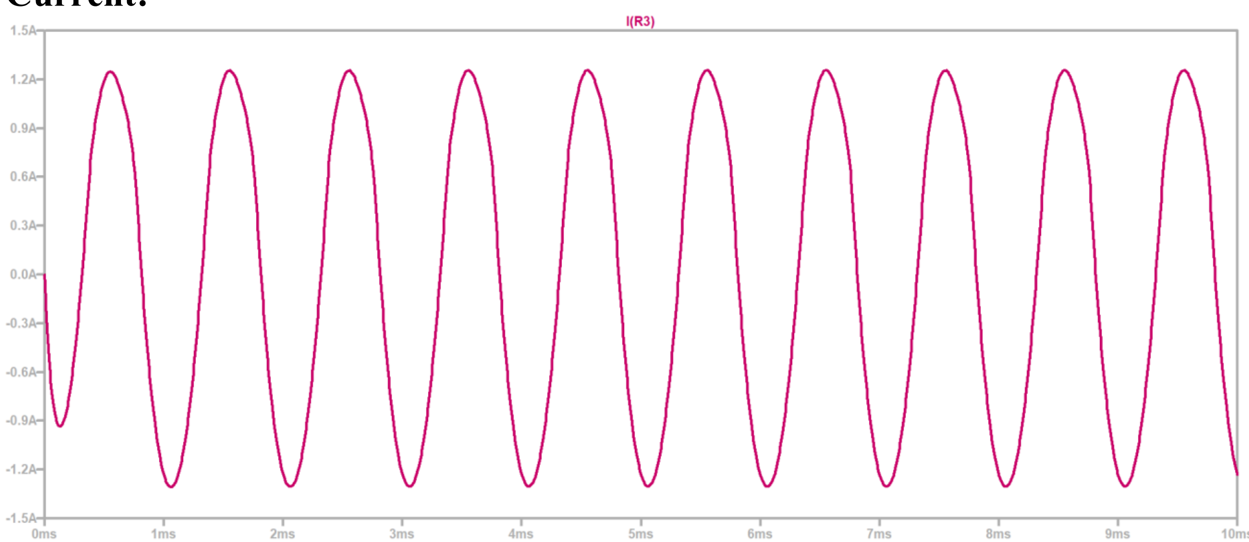
Input:



Output:



Current:



Short Conclusion: the PNP driver and Class-AB output pair operate together as a single, stable power stage. This final combined stage defines the load requirements for the earlier parts of the amplifier and ensures reliable high-power audio output.

Mosfet PreAmplifier:

Purpose of the stage:

- To amplify the millivolt microphone signal
- To provide high input impedance
- Biasing calculations (ID, VGS, RD, RS)
- AC gain calculation: Av = gm × RD
- Input and output impedance

Calculations:

For M2 2N7000 :-

Assume $I_D = 2mA$

Voltage drop across drain.

$V_D = 17V$

$$R_D = R_{17} = \frac{24-17}{2m}$$

$$\boxed{R_{17} = 3500\Omega}$$

$V_S = 3.024V$

$$R_{20} + R_{21} = R_S = \frac{3.024 - 0}{2m}$$

$$\boxed{R_S = 1512}$$

$I_G = 0A$

Voltage drop across gate.

$V_G = 7.92V$

$$R_{B1} = R_{tot} \cdot \frac{V_G}{V_{CC}} = 120k \cdot \frac{7.92V}{24}$$

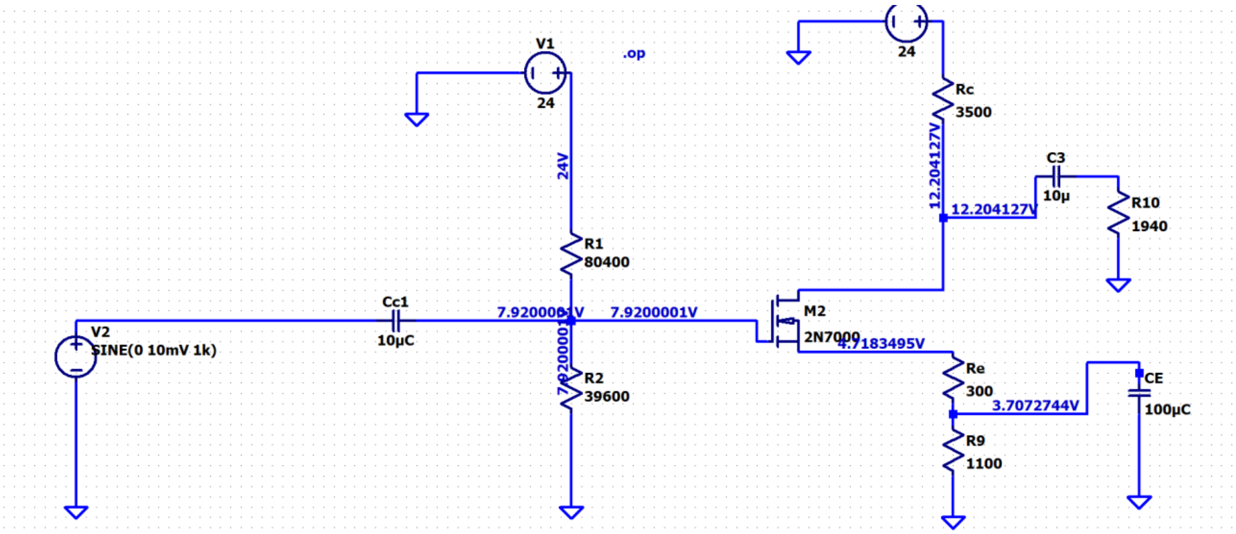
$$\boxed{R_{19} = 39600\Omega}$$

$$R_{tot} = \frac{V_{CC}}{I_{di}} = \frac{24}{200 \times 10^{-6}} = 120000\Omega$$

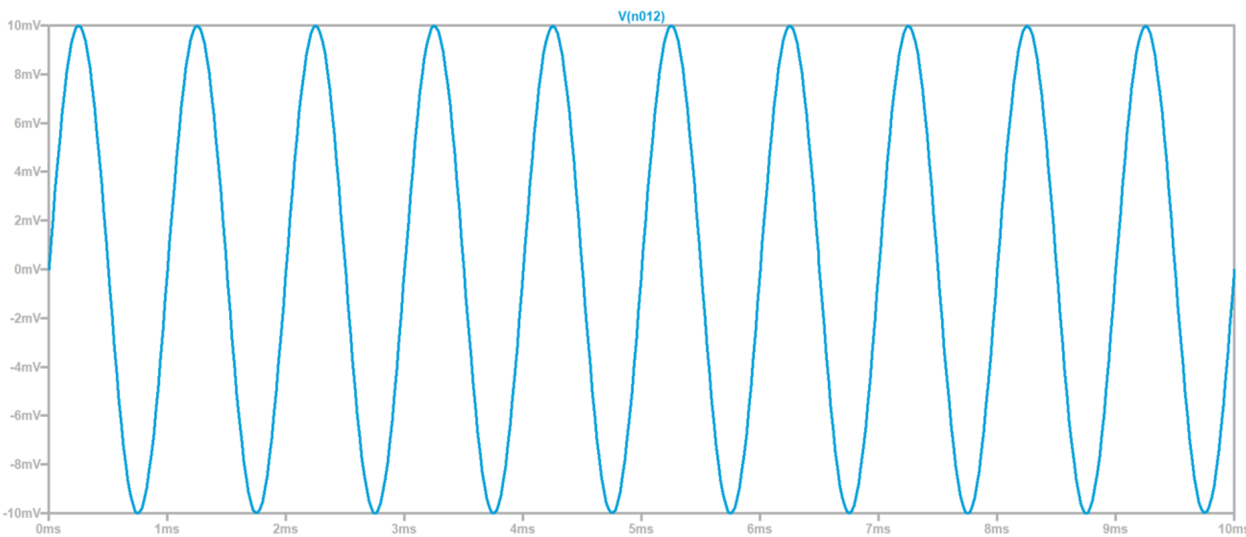
$$R_{B1} = R_{18} = R_{TOT} - R_B = 120000 - 39600$$

$$\boxed{R_{18} = 80400\Omega}$$

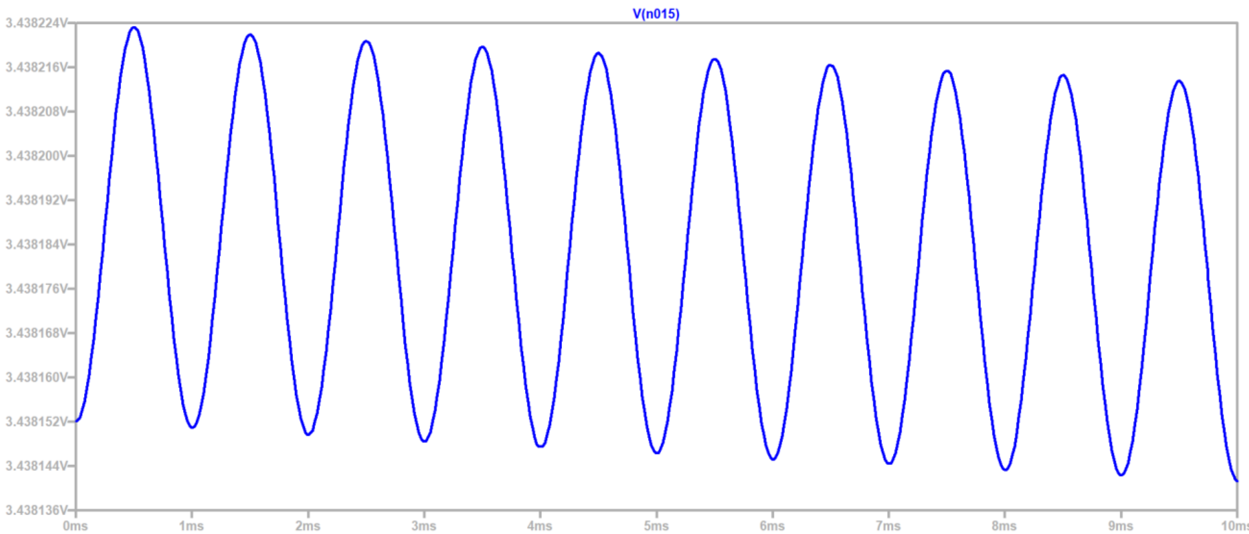
Circuit Diagram (LTspice Schematic):



Input:



Output:



BJT PreAmplifier:
Reason for adding BJT stage:

- Higher gain
- Better voltage amplification
- Bias calculations (IB, IC, VCE)
- AC gain calculation
- Frequency response

Calculations:

For Q5 BJT:-

$R_L = 1559.85\,\Omega$

Assume $i_c = 3.3mA$

Assume Drop at collector is half.

$V_c = \frac{V_{cc}}{2}$

$V_c = \frac{24}{2}$

$V_c = 12V$

$R_C = R_{12} = \frac{24-12}{3.3\times10^{-3}} = 3636.3\,\Omega$

$\boxed{R_{12} \simeq 4k}$

$i_B = \frac{i_c}{\beta}$

$\therefore \beta = 100$

$i_B = \frac{3.3\times10^{-3}}{100}$

$\boxed{i_B = 3.3\times10^{-5}A}$

10 to 12 times of i_B = $\boxed{5\times10^{-4}A}$

$V_B = V_{BE} + V_E$

$V_B = 0.7 + 2.4$

$\boxed{V_B = 3.1V}$

$$R_{B1} = R_{13} = \frac{24 - 3.1}{5\times10^{-4}}$$

$$\boxed{R_{13} = 41800\,\Omega}$$

$$R_{B2} = R_{14} = \frac{3.1 - 0}{5\times10^{-4}}$$

$$\boxed{R_{14} = 6200\,\Omega}$$

$$A_v = \frac{R_C \parallel R_L}{r_e + R_E}$$

$$r_e = \frac{V_T}{I_c} = \frac{25m}{3.3m}$$

$$r_e = 7.57\,\Omega$$

$$50 = \frac{4k \parallel 1559}{7.57 + R_E}$$

$$\boxed{R_E = 18\,\Omega} \text{ (unbypassed)}$$

$$R_{in} = R_{B1} \parallel R_{B2} \parallel (\beta+1) R_E(\text{unbypassed})$$

$$\therefore \beta = 100$$

$$R_{in} = R_{13} \parallel R_{14} \parallel (101)18$$

$$R_{in} = \frac{41800 \times 6200}{41800 + 6200} \parallel (101)(18)$$

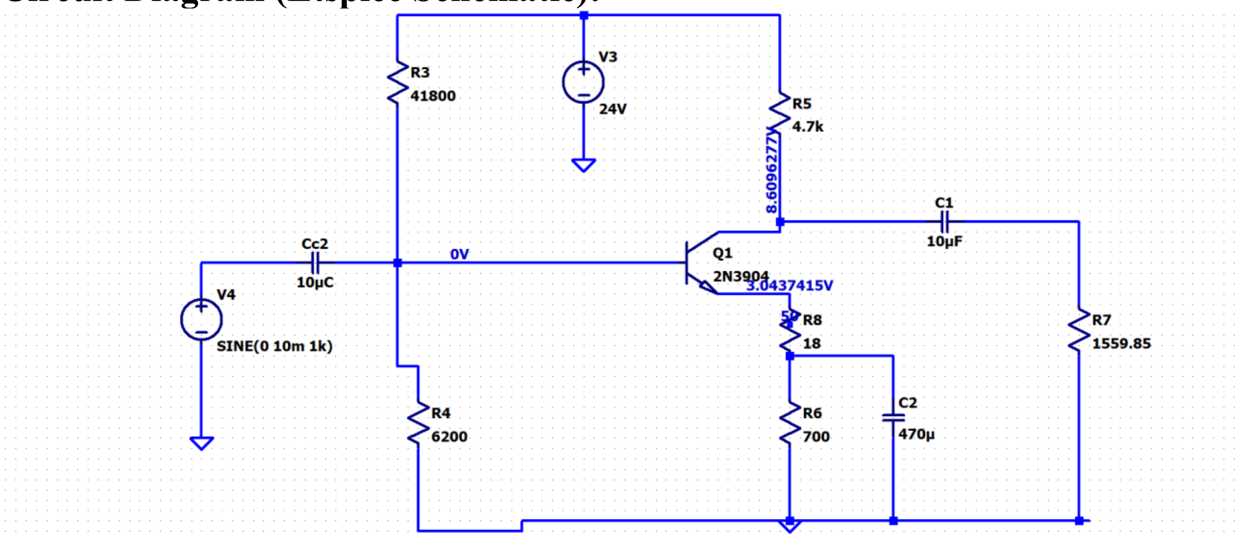
$$R_{in} = 5399.1 \parallel 1818$$

$$R_{in} = \frac{5399.1 \times 1818}{5399.1 + 1818}$$

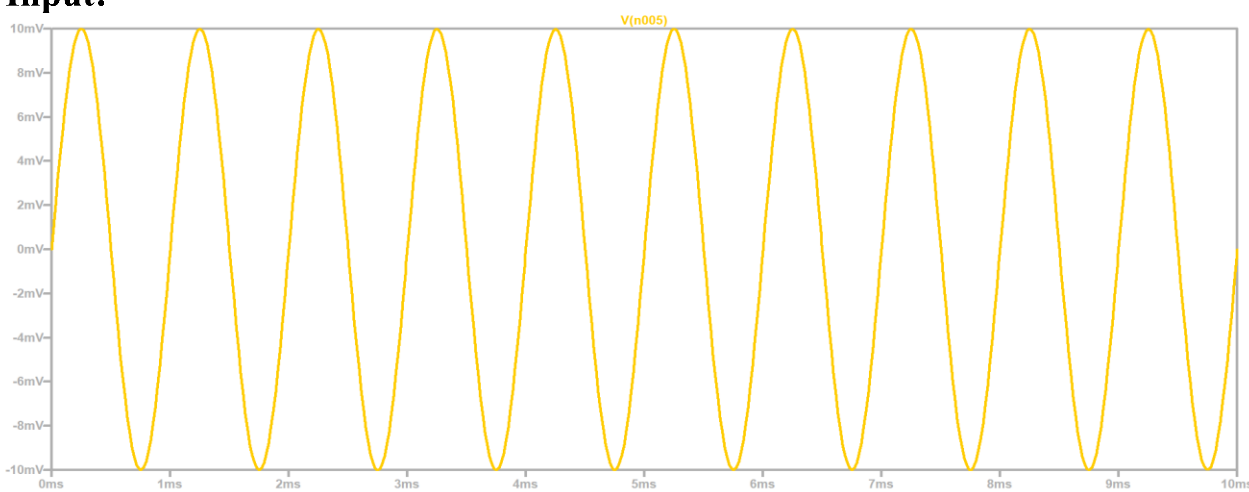
$$R_{in} = 1360.04\,\Omega$$

$\Rightarrow R_L$ for BJT $= 1360.04\,\Omega$

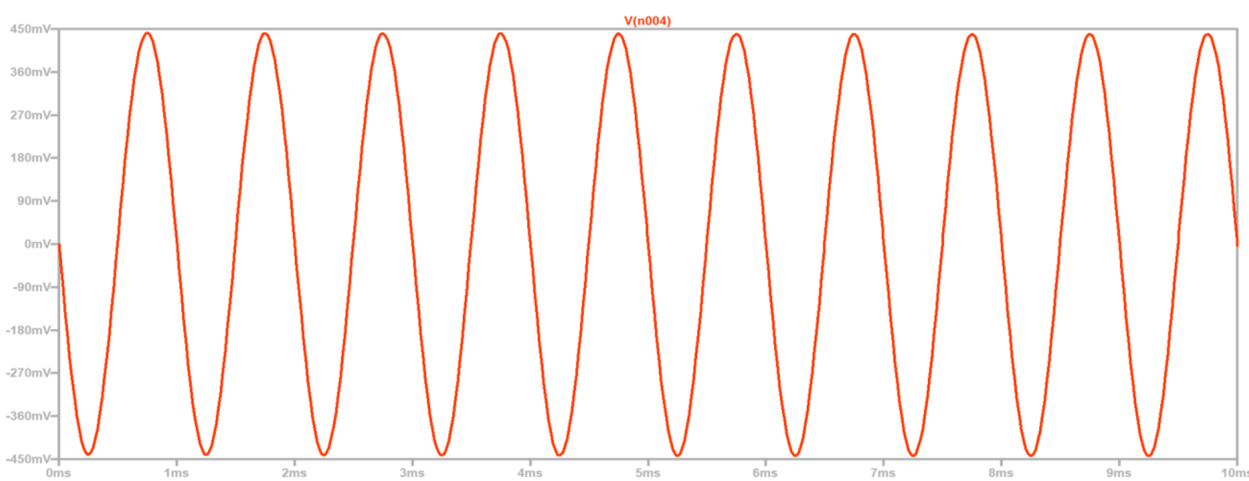
Circuit Diagram (Ltspice Schematic):



Input:



Output:



Combined PreAmplifier (MOSFET + BJT):

Reason for combining both stages

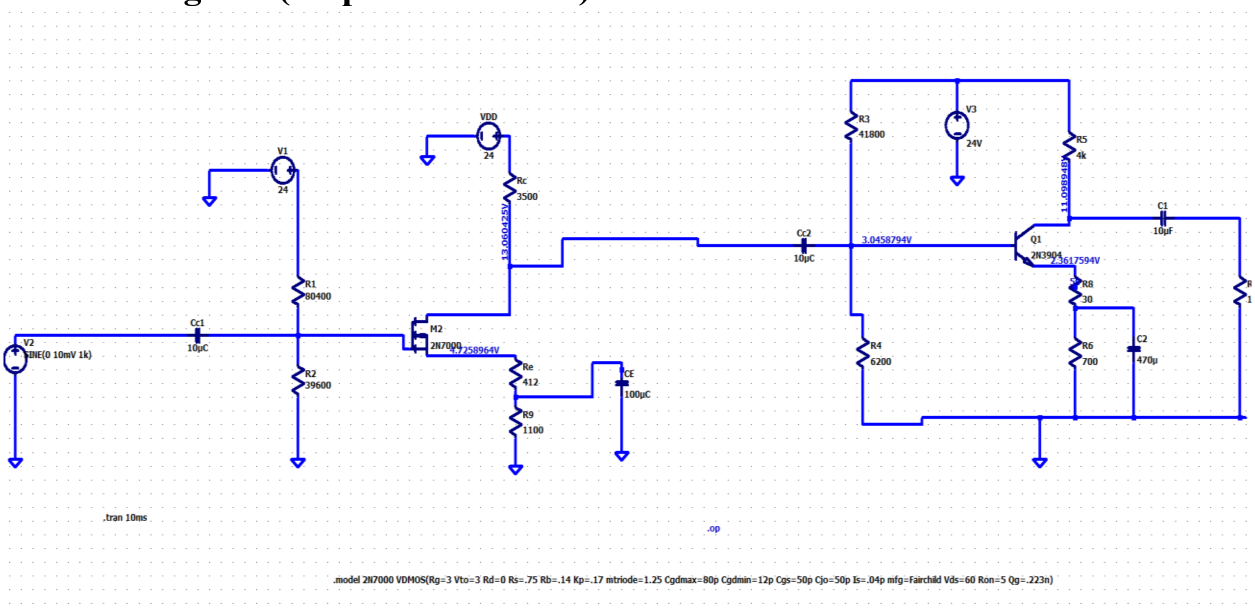
MOSFET → high input impedance
BJT → high voltage gain

Feedback Network Design

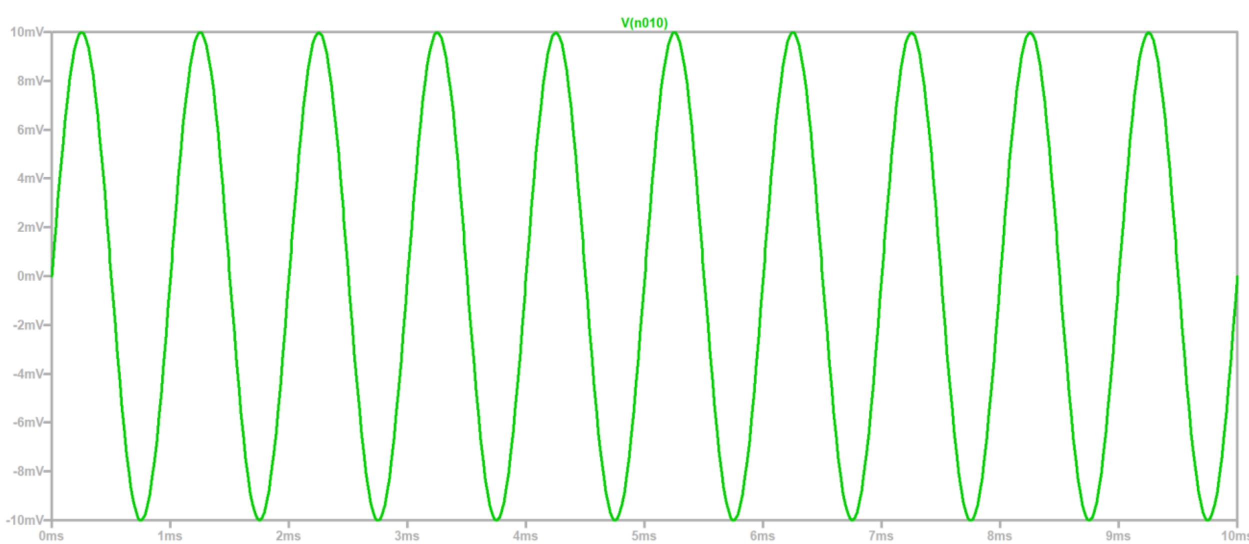
Gain after feedback (Av ≈ 100)
Stability improvement
Reduced distortion

Calculations:

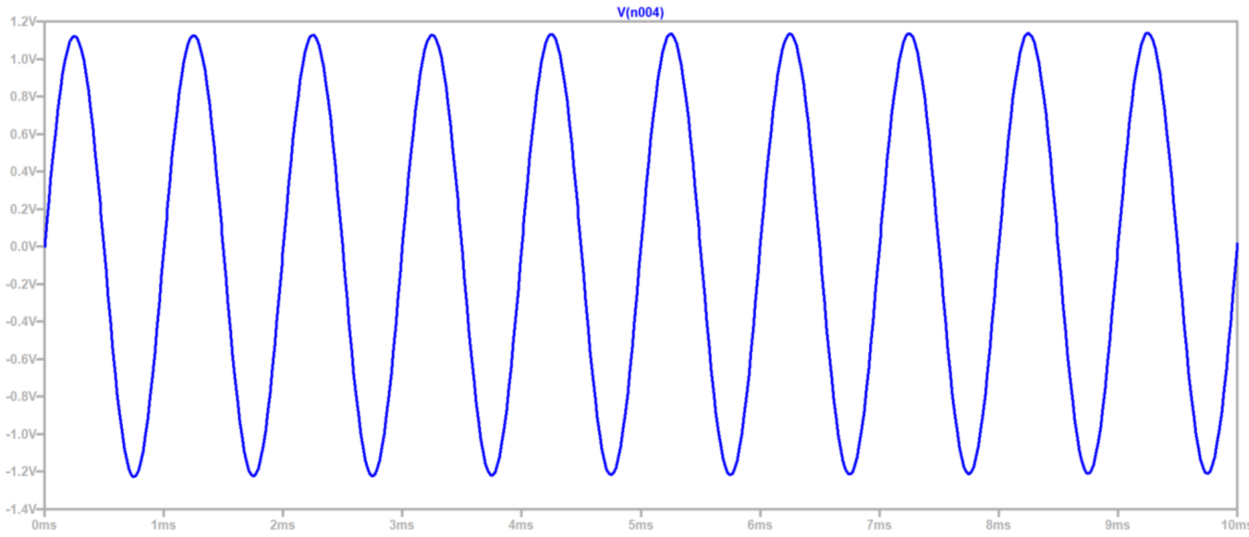
Circuit Diagram (Ltspice Schematic):



Input:



Output:

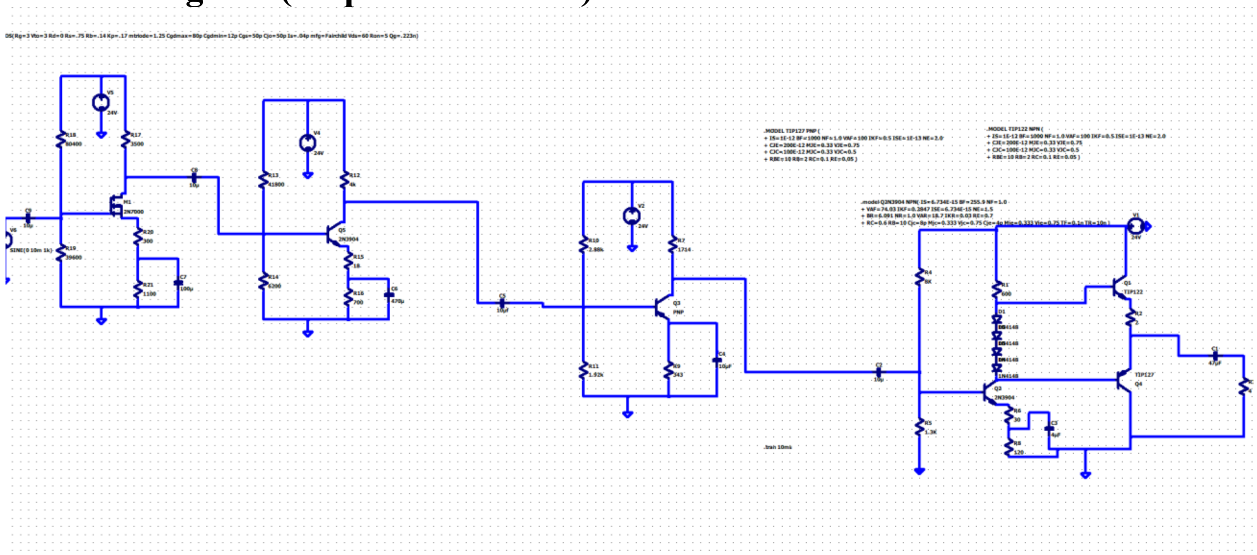


Complete Audio Amplifier:

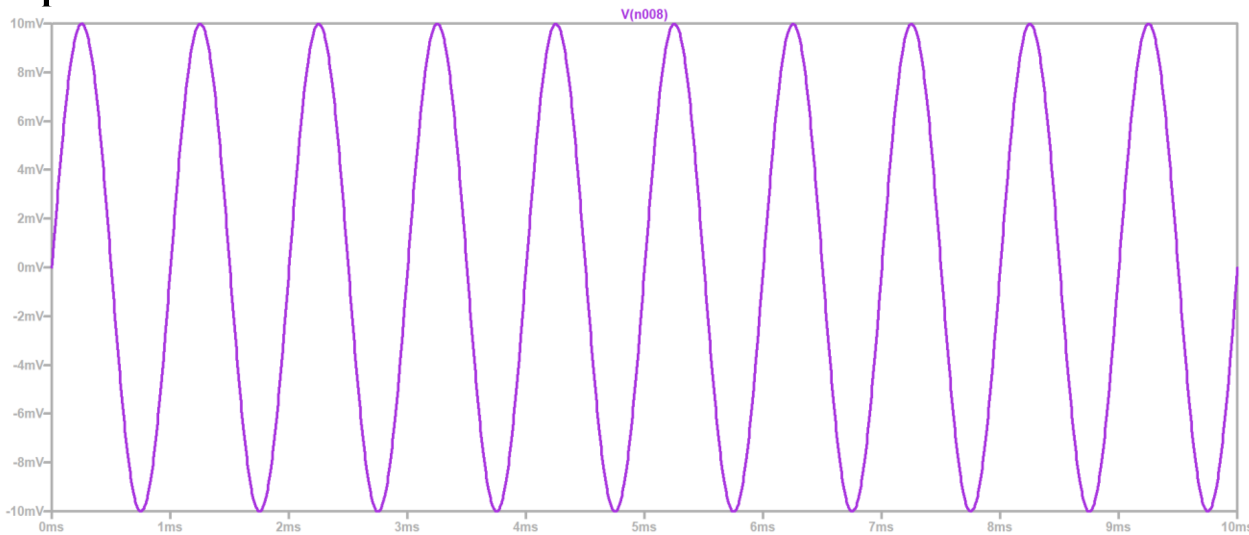
After designing and testing each stage separately, all blocks—pre-amplifier, PNP driver, and Class-AB power amplifier—were connected together to form the complete audio amplifier. In the final combined circuit, the pre-amplifier provides the required voltage gain and high input impedance for the microphone, the PNP driver boosts the current to properly drive the output stage, and the Class-AB amplifier delivers the final 3–5 W power to the 8-Ω speaker.

When all stages were connected, the full amplifier was tested in LTspice. The simulation showed smooth signal flow from the input to the output, correct biasing across all transistors, clean voltage gain, and a stable frequency response. The final output waveform was undistorted at the maximum swing, and the amplifier successfully met the required power and bandwidth specifications.

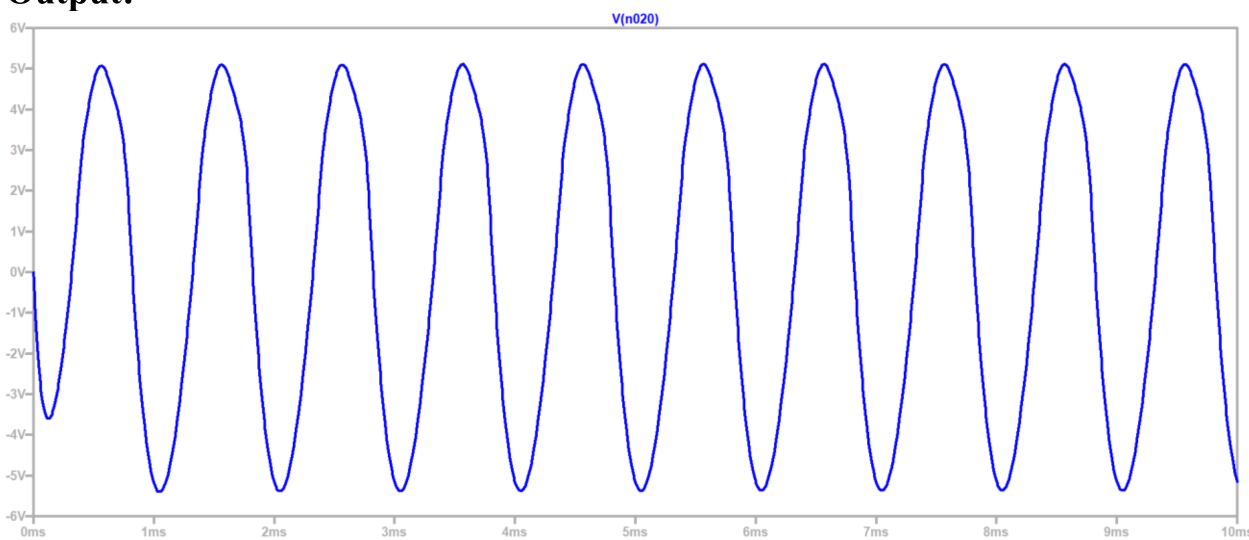
Circuit Diagram (Ltspice Schematic):



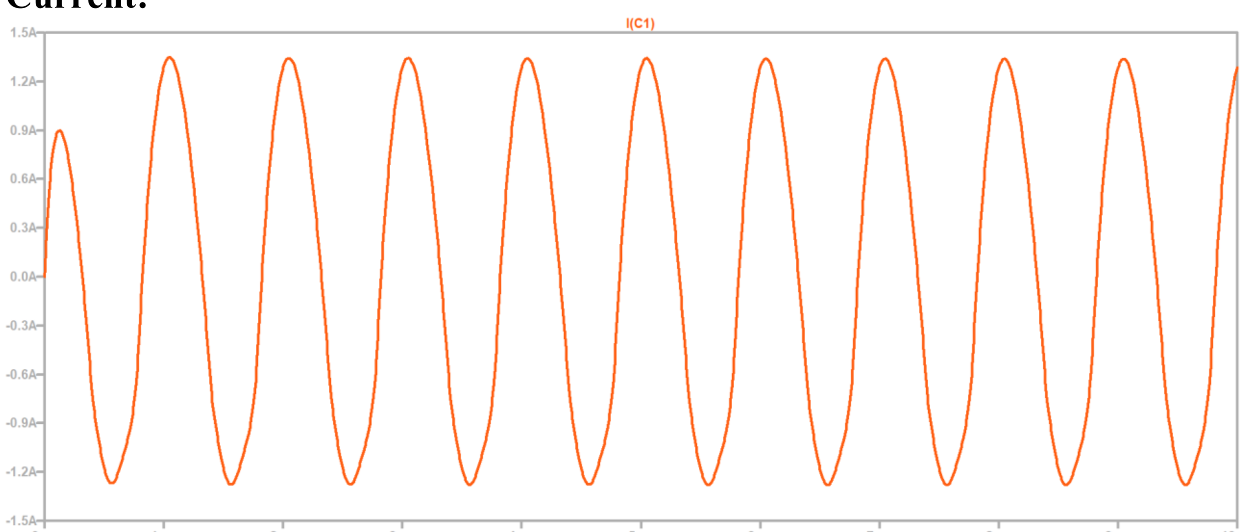
Input:



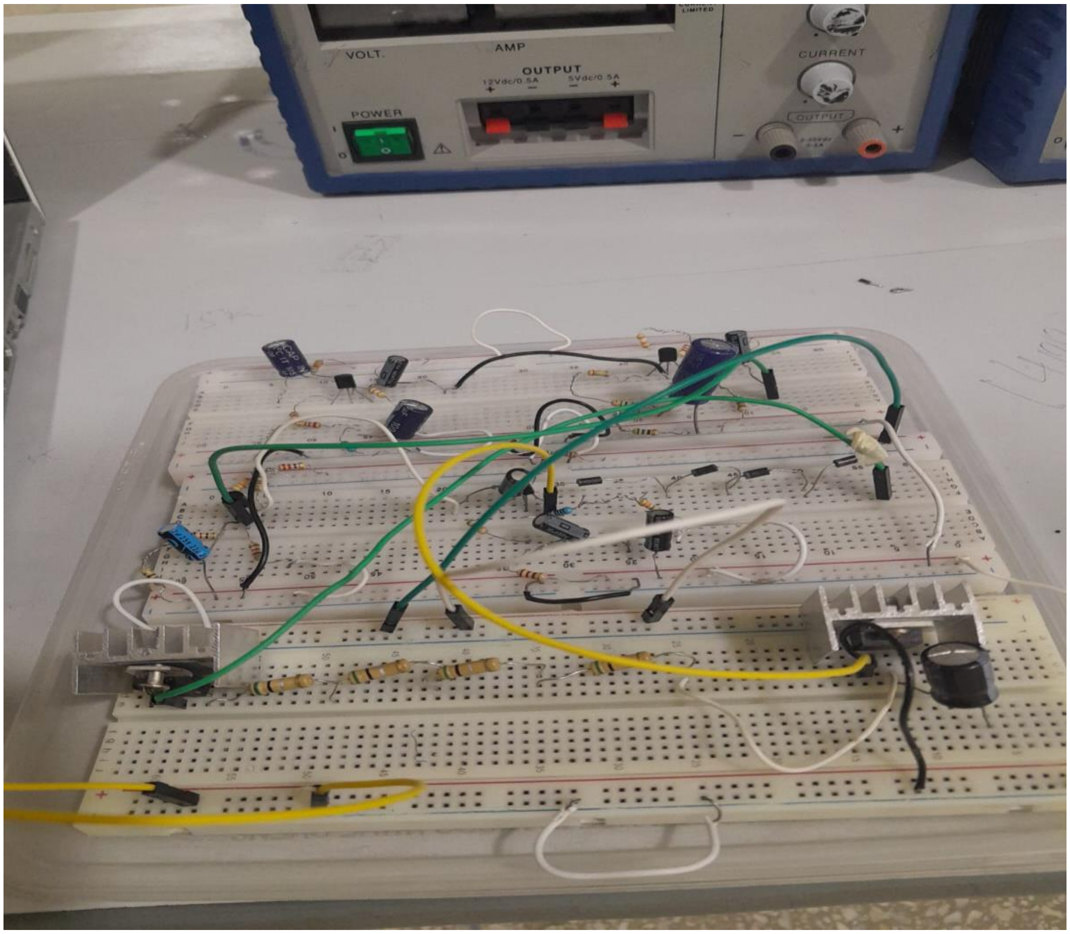
Output:



Current:



Hardware:



Conclusion:

In this project, a complete discrete-component audio amplifier was successfully designed, analyzed, and simulated. The design began from the Class-AB output stage to ensure the required 3–5 W power delivery to an 8-Ω speaker. The PNP driver stage was then developed to provide the necessary drive current and was later combined with the output transistors to form a stable and efficient power stage. After that, impedance levels were matched, and a pre-amplifier was designed to supply the correct input gain and signal conditioning for the power section. Each stage was first calculated individually and then verified through LTspice simulations. Finally, all stages were connected to form one complete audio amplifier. The final simulation confirmed smooth signal flow, proper biasing, wide bandwidth, and clean undistorted output at maximum swing. The amplifier met all design specifications for gain, frequency response, and output power.

Overall, the project demonstrated a complete understanding of multi-stage amplifier design, including biasing, impedance matching, power calculations, and practical circuit integration.